Engineering Mechanics 1 · Unit_03 · UNIT 3

DRAFT — UE26ME142A · Unit_03 · UNIT 3 · CLASS 33 Introduction to Centroid

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

CLASS 33 Introduction to Centroid

1 CLASS 33 Introduction to Centroid

Study focus: CLASS 33 Introduction to Centroid. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000002 · EV0000003

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

If you consider any external force in nature, each and every external force is a distributed force because external forces are applied mechanically on a body and they will have a definite area of contact.

Key details

- However small the area might be.
- That's why each and every external force is a distributed force.
- Actually it's not a point, it's the area of the tyre which is in contact with the pavement.
- You can see in this enlarged weave.
- But actually, it is a distributed force as shown in this figure.
- We are going to assume the same force is a concentrated force because the area of the contact between the tire and the surface or the pavement is very, very small compared to the other dimensions of the car.
- It is a concentrated force by considering the resultant of this distributed force.
- These are the few examples of the distributed forces.
- And in actual sense, concentrated forces doesn't exist in nature.
- Please remember this point.
- Study about the earth's gravitational attraction.
- In this line, I am going to explain you how to locate the center of gravity of any three-dimensional body practically.
- Body, three dimensional body of any size and shape.
- Let M be the mass of this body.
- Through the point A, the body is suspended with the help of this card.

- If I draw the free body diagram of this body, there are only two external forces acting on the body.
- The first one is tension in the card, and the one more is the self-weight of this body.
- This body is in equilibrium.
- These two external forces, one is tension and one more is self-weight.
- These two should be equal, opposite, and collinear.
- That's why this W is along the same line as that of this tension in the card, and it is downward because the tension in the card is upward.
- This is the case when the body is suspended through point A.
- Again, we are going to draw the free body diagram of this body.
- There are only two external forces.
- One is tension in the card and one more is the self rate.
- The body is in equilibrium, so these two forces should be equal, opposite and collinear, then only they'll get cancel each other, then the body will be in a state of equilibrium.
- That's how we are going to get the line of action of this.

Example or application from the lecture

- I'm going to consider the example of this automobile wheel, which is in contact with the pavement as shown in this figure.
- If you remember the numericals of any automobile example, like a car, when we consider the free body diagram of a car, we are going to consider the reaction from the pavement onto the wheel like this, as a concentrated force.
- Example, when we consider the distance between the two wheels, the area of this contact is very small.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000002 · EV0000003 · EV0000004 · +1 source units in pack.json

Core explanation · 05:00–09:59

W It will be along the same line as that of the card, but in the opposite direction.

Key details

- And this also we are going to mark by drilling a small hypothetical hole on this rigid body.
- If you observe the line of action for the various cases, all the lines of action are concurrent at a particular point.
- That point we call it as the center of gravity.
- Irrespective of the orientation of the body this point will be remains the same.
- Then how to locate determine the center of gravity the same g mathematically.
- So to locate the center of gravity we are going to apply the concept of method of moments.
- Let g be the center of gravity of this body.
- Then the coordinates of this g will be always x bar, comma, y bar, comma, z bar.
- To find the value of X bar Y bar Z bar mathematically, we are going to apply the principle of moments.

- For this, what we are going to assume is we know each and every body will be having infinite number of.
- Each and every particle will be acted upon by the earth's gravitational force.
- Downward and the magnitude of that force is DW.
- Since the particle is very small,.
- It's just like a point, and the coordinates of this point are x, y, comma, z.
- According to this statement of principle of moments.
- The moment of the resulting gravitational force W about any axis it can be X, Y, R Z, will be always equal to.
- Some of the moments of the gravitational forces acting on all these particles about the same axis.
- This is the standard statement of the principle of moments.
- Then how to find the resultant gravitational force here.
- The resultant gravitational force W will be equal to sum of the weights of all these particles within the body.
- Next, moment of the.
- This particle about x axis.
- The moment of this particle about x axis will be equal to force into perpendicular distance.
- What is force?
- And distance is y.
- This is the moment of only one particle.
- Next we want the sum of the moments.
- Moment of the resulting gravitational force that is LHs is forces w into perpendicular distances y bar.
- Once we simplify we are going to get the value of y bar.
- Capital W And this W is total weight of the body.
- We got the value of this y bar.
- next we know the weight of the body will be always equals to mass into gravity.
- Substitute the value of w and dw here, we are going to get the one more expression for x bar, y bar and z bar.
- This is independent of the gravity.

Example or application from the lecture

- Here, let's consider there is a three-dimensional body of any size and shape.
- Let's apply here the moment of the resultant gravitational force W.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000006 · EV0000007 · EV0000008 · +1 source units in pack.json

Core explanation · 10:00–14:59

This is the point through which we assume the entire mass of the body is concentrated.

Key details

- This is called as center of mass.

- X bar, y bar, and z bar gives us the coordinates of the center of mass.
- The center of mass and center of gravity will coincide as long as the acceleration due to gravity is constant throughout the body, and at the same time, the forces are parallel to each other.
- Next, we know dense mass is equal to density into volume.
- And volume also we know area into thickness.
- Uniform density throughout as well as a uniform thickness throughout.
- In that case in both numerator and denominator, we are going to get.
- Density rho as constant as well as the thickness also as constant.
- coordinates of the centroid of the lamina.
- And centroid is the point where we assume the entire area of the body is concentrated.
- We are going this is the general formula to look at the centroid x bar integral of x into dA by total area, y bar integral of y into d a by total area.
- But when we are.
- considering any elemental strip within that lamina, then instead of considering the general expression, we are going to be more specific.
- if you look at this figure, there is a lamina something like this.
- Then I want to locate the centroid of this lamina.
- strip within this lamina.
- Then the coordinates of the centroid of this.
- Point R XC, Y C.
- The formula is going to be X bar is equals to integral of X C into DA by total area.
- Y bar is integral of.
- Yc into DA by total area.
- Next, there is one more formula.
- That's why this card is a first movement of.
- Next, there is one more formula.
- X bar is equals to summation of AI Xi by total area.
- Y bar is summation of AI Yi by total area.
- We are going to make use of.
- This particular formula in the composite areas.
- Composite area is nothing but any lamina figure which will be having some basic figures like combination of basic figures like.
- triangle, rectangle, square, circle, semicircle like that.
- And we can write it as a1 x1 plus a2 x2 plus so on.
- Plus a nxn by total area.
- That's how we are going to expand this particular formula.
- Next is axis of symmetry.
- You can look at this.
- Here there is a right circular cone.
- If you draw vertical line passing through the apex here through this point.
- So the center of gravity R centroid will be on this because.
- The moment of left of this will be equals to moment of right of this, these two will get cancel each other.

- It is applicable even for laminus as well.

Example or application from the lecture

- In that case, I am going to consider this elemental strip.
- We are going to make use of the concept of first moment of area to locate the center of a given lamina.
- Axis of symmetry is any line at the axis which divides the given lamina or a figure at any three-dimensional body into two equality.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000010 · EV0000011 · EV0000012 · +1 source units in pack.json

Core explanation · 15:00–19:59

This is one more figure.

Key details

- If we draw a line passing through this, this right of the section is the mirror image of the left of the section.
- Then capital A is the area of this triangle.
- Let that be equals to A is equals to B into H.
- Next, I need to find the Y coordinate of the centrate.
- This is before that this is the reference axis, let it be x axis.
- The width, the depth of this element need to be dy.
- This element can be anywhere from this to this point T.
- And the depth of the element dy is also a variable.
- It can be.
- Anything from 0 to H.
- Next, the width of the element is also a variable because when the value of y is equals to 0, the width of this element will be equal to b.
- As the value of y increases, the width of the element decreases.
- That's why I'm going to call it as x.
- To find the y bar we have a formula y bar is equal to integral of yc into d a by total area e.
- DA is area of this element.
- The area of the element, since it is in the form of rectangle, almost like a rectangle, d A will be equal to B into H.
- Here B is X into depth of the element is DY.
- Y C we should find.
- The what is YC?
- This shaded area will have a centrite C.
- When Dy is very small, this Yc is approximately equal to Y.
- This will be equal to integrate integral of y.
- Yc is y into d is x into dy divided by total area capital E.
- X is a variable and y is also variable.
- Because we have the term dy here.

- This is x.
- And this we know b, and this we know y.
- And the total height of the rectangle is h.
- This is also known to us.
- We can use the concept of similar triangles because these two are parallel to each other.
- By using the concept of similar triangles,.
- Base of the smaller triangle that is x by base of the bigger triangle that is b will be equals to height of the smaller triangle.
- The height of the smaller triangle is only this.
- This will be equals to.
- Total h minus y.

Example or application from the lecture

- Next, let's find the y coordinate of central of a triangle of base B and a height H from the method of integration.
- You can consider either a right angle triangle or just any triangle, I'm going to consider a triangle like this.
- Let us consider a triangle A, B, C, Let B be the base of this triangle, let H be the height of the triangle.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000014 · EV0000015 · EV0000016 · +1 source units in pack.json

Core explanation · 20:00–24:59

x in the form of constants b and h and a variable y.

Key details

- I'll take 1 by s here integral of.
- Y into X is B into H minus Y by H into D Y.
- This is 1 by A.
- As I said before, B and H are constants, so you can take them out.
- B and H.
- Y is varying from 0 to H.
- Then in Y into H is H into 1.
- Minus y square into d y d by outside the bracket.
- Simplify this, this will be equals to area is b into h by half into base into height.
- This is y cube by three zero two h.
- Next, this will be equals to.
- Two H to by B will get canceled here H square into.
- Once we substitute the value of y as h, so h square minus of h cube by 3, and now the next term is going to be 0.
- This is going to be H cube by six.
- This will be equal to two three.
- One by three H.

- This is the Y coordinate of the central of a triangle with respect to bottom plus member.
- This is x-axis passing to the bottom most member this is where the centroid will be, and this is y bar that is h by three.
- If the axis is passing to the topmost member, x-axis is passing to the topmost member at the apex of the triangle.
- In that case, the y-coordinate of the central is 2h by 3.
- That's it for today's class.

Example or application from the lecture

- This is a triangle if we consider.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000018 · EV0000019 · EV0000020 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
CLASS 33 Introduction to Centroid transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/033_CLASS_33_Introduction_to_Centroid__67dd96fa80cc7e7c.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 33 Introduction to Centroid transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/033_CLASS_33_Introduction_to_Centroid__67dd96fa80cc7e7c.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 33 Introduction to Centroid transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/033_CLASS_33_Introduction_to_Centroid__67dd96fa80cc7e7c.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
CLASS 33 Introduction to Centroid transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/033_CLASS_33_Introduction_to_Centroid__67dd96fa80cc7e7c.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 15 rendered pages in 033_Centroid__9f0e55503970de0e.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.md have no readable `.slides.md` text; their original page images are included.
3. 3 rendered pages in 039_UE24CV141A__Unit_3_Beams__9fc34bbd5ca57988.slides.md have no readable `.slides.md` text; their original page images are included.
4. 3 rendered pages in 039_UE25CV141A_Unit_3_Class_48_Lecture_PPT__4d28f283532e284e.slides.md have no readable `.slides.md` text; their original page images are included.
5. 3 rendered pages in 042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.slides.md have no readable `.slides.md` text; their original page images are included.
6. 5 rendered pages in 043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.slides.md have no readable `.slides.md` text; their original page images are included.
7. 6 rendered pages in 044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.slides.md have no readable `.slides.md` text; their original page images are included.
8. 4 rendered pages in 044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.slides.md have no readable `.slides.md` text; their original page images are included.
9. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
10. 6 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
11. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.

12. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
13. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
14. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
15. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00002	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0065. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00003	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0065. Sidecar variants compare...
SRC00004	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0065. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00005	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0065. Sidecar variants compare...

C Revision and retrieval practice